Project 1: AES Encoder & Decoder

Introduction

The Advanced Encryption Standard (AES) is a symmetric-key encryption algorithm used to secure data in communication systems, storage devices, and embedded applications. Software implementations on CPUs often suffer from sequential execution and limited throughput. In contrast, hardware accelerators can exploit parallelism and pipelining to perform encryption and decryption much faster and with lower energy consumption. This makes dedicated AES hardware essential for high-performance and security-sensitive systems.

Project Description

In this project, you will design and implement a hardware accelerator for AES encryption and decryption. You will build the AES datapath and control logic in SystemVerilog, implementing all major stages, key expansion, SubBytes, ShiftRows, MixColumns, and AddRoundKey. You will optimize the architecture for speed, area, and power, using techniques such as pipelining or loop unrolling to increase throughput. You will verify functionality through simulation and evaluate the design on an FPGA.

Expected Results

You will deliver a working AES encoder–decoder hardware accelerator that performs real-time encryption and decryption. The design should show significantly higher throughput and lower latency than the CPU-based implementation, demonstrating the benefits of custom hardware for secure and efficient data processing.

Useful References

1. Advanced Encryption Standard Wikipedia
2. FPGA Implementation of Advanced Encryption Standard Algorithm
3. Compact and Efficient Encryption/Decryption Module for FPGA Implementation of the AES Rijndael Very Well Suited for Small Embedded Applications